11.7 Simpson's Rule in a Rod

WORKSHEET

The question

STEP II 1989, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Two systems have the same mass, the same centre of mass, and the same moment of inertia about every axis through it. Why does the parallel axis theorem then give the same moment of inertia about every other axis too?

STAGE 1 The rod

Ask yourself Put masses p at each end and q at the middle. By symmetry the centre of mass is right. What do the total mass and the moment of inertia about a perpendicular axis through the centre require?

Ask yourself Does this work for *every* axis through the centre, not just the perpendicular one?

STAGE 2 The rectangle

Ask yourself Cut the lamina into thin strips parallel to one side. Replace each strip by its three equivalent particles. What do you get, and what can you do next?

STAGE 4 A non-uniform rod

Ask yourself Let the rod have length L and density $\frac{2mx}{L^2}$ at distance x from one end. Put masses p_0, p_1, p_2 at $x = 0$, $\frac{L}{2}$, L. Matching the total mass, the centre of mass and the moment of inertia about the end gives three equations. What are they?

What's really going on

Simpson's rule

Matching moments of a mass distribution is the same problem as designing a numerical integration rule. A rule $\int_{-a}^{a} f(x)\,\mathrm{d}x \approx w_0 f(-a) + w_1 f(0) + w_2 f(a)$ that is exact for $f = 1, x, x^2$ must satisfy

$$w_0 + w_1 + w_2 = 2a, \qquad -aw_0 + aw_2 = 0, \qquad a^2 w_0 + a^2 w_2 = \tfrac{2}{3}a^3,$$

exactly the conditions for the rod (with $f = 1, x, x^2$ playing the roles of mass, first moment and moment of inertia). The solution is $w = 2a(\frac{1}{6}, \frac{2}{3}, \frac{1}{6})$: Simpson's rule. A bonus comes from symmetry: the rule is also exact for $f = x^3$, since both sides vanish, so Simpson's rule integrates cubics exactly, and its error for smooth functions is proportional to the fourth derivative. A rod therefore matches its three particles not just in mass, centre and moment of inertia, but in the third moment too.

Gaussian quadrature and the inertia tensor

If the positions as well as the masses are free, two particles suffice: half the mass at each of the points $\pm\frac{a}{\sqrt{3}}$. Check: $2\cdot\frac{1}{2}m\cdot\frac{a^2}{3}=\frac{1}{3}ma^2$. This is the two-point *Gauss–Legendre rule*, $\int_{-a}^{a} f \approx a\big(f(-\frac{a}{\sqrt{3}})+f(\frac{a}{\sqrt{3}})\big)$, which is also exact for cubics. Carl Friedrich Gauss showed in 1814 that n cleverly placed points can integrate polynomials of degree $2n-1$ exactly, and Gaussian quadrature is now a standard method of numerical analysis.

In mechanics, the moments of inertia of a body about all axes through its centre are encoded in a symmetric 3×3 matrix, the *inertia tensor*. Two bodies with the same mass, centre of mass and inertia tensor respond identically to all forces, which is why engineers can replace a complicated part by a few point masses in a simulation. The eigenvectors of the inertia tensor are the *principal axes*, the axes about which a body can spin steadily.

Try this next

1. **Simpson's rule.** Use Simpson's rule with a single interval to estimate $\int_0^1 \frac{\mathrm{d}x}{1+x^2}$, and compare with $\frac{\pi}{4}$. Check that the rule is exact for $\int_{-1}^{1} x^3\,\mathrm{d}x$ and $\int_{-1}^{1} x^2\,\mathrm{d}x$ but not for $\int_{-1}^{1} x^4\,\mathrm{d}x$.

2. **All axes alike.** Show that the moment of inertia of a square lamina about any axis through its centre in its plane is $\frac{1}{3}Ma^2$. (*Hint:* the nine equivalent particles are symmetric under rotation by 90°; what does that say about a quadratic expression in $\cos\psi$ and $\sin\psi$?)

3. **A triangle.** Show that a uniform triangular lamina of mass M is dynamically equivalent to three particles of mass $\frac{1}{3}M$ at the midpoints of its sides.

4. **A solid cube.** Find a dynamically equivalent system of 27 particles for a uniform solid cube, and use it to find the moment of inertia of a cube of side $2a$ about an edge.

Further reading

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), the chapter on rigid bodies and moments of inertia.
- **Endre Süli & David Mayers, *An Introduction to Numerical Analysis*** (Cambridge University Press), the chapters on numerical integration, including Simpson's rule and Gaussian quadrature.
- **Philip Davis & Philip Rabinowitz, *Methods of Numerical Integration*** (Dover). The classic reference.

The full guided solution is in Section 11.7:
https://beyondstep.keithmcnulty.org/ch11/simpsons-rod.html